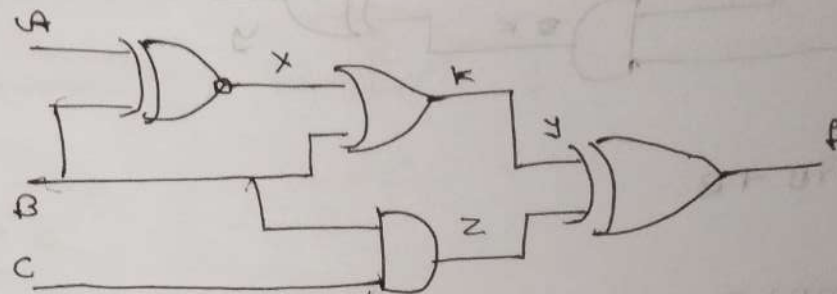


Assignment - 2

1) Derive the output expression f for the below circuit and draw the simplified circuit



$$X = \overline{A}B + AB$$

$$K = \overline{A}B + AB + B$$

$$\overline{A}B + B$$

$$K = \underline{\underline{\overline{A} + B}}$$

$$Z = B.C$$

$$f = K \oplus Z$$

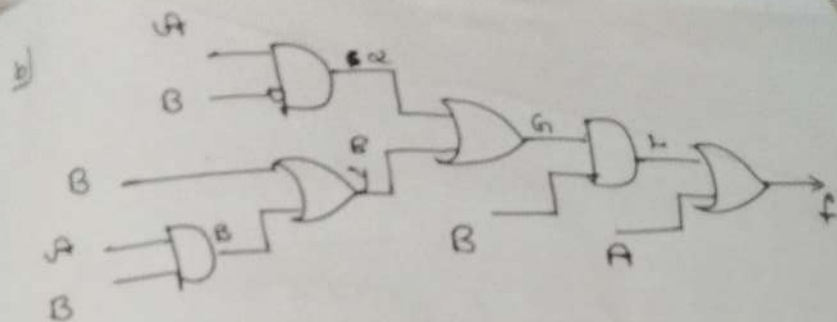
$$\overline{A} + B \oplus BC$$

$$= (\overline{A} + B).BC + (\overline{A} + B)(\overline{B.C})$$

$$\overline{A}.B.BC + (\overline{A} + B)(\overline{B} + \overline{C})$$

$$\cancel{\overline{A}.B.BC} + \overline{A}\overline{B} + \overline{A}\overline{C} + B\overline{B} + B\overline{C}$$

$$\underline{\underline{\overline{A}\overline{B} + \overline{A}\overline{C} + B\overline{C}}}$$



$$\alpha = A \cdot B$$

$$\gamma = A \cdot B + B$$

$$\Rightarrow B$$

$$G_1 = \alpha + \gamma$$

$$G_1 = A \cdot B + B$$

$$G_1 = (A + B)$$

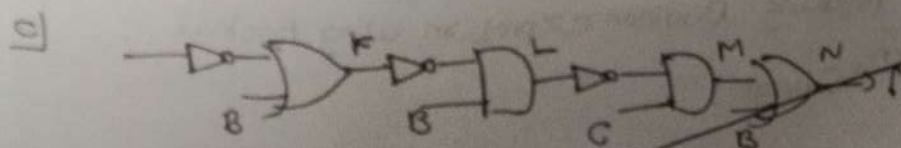
$$L = (A + B) \cdot B$$

$$L = AB + B \cdot B$$

$$L = AB + B$$

$$L = B$$

$$f = A + B$$



$$K = A + B$$

$$L = (A + B) \cdot B$$

$$L = (A + B) \cdot B = AB + B \cdot B$$

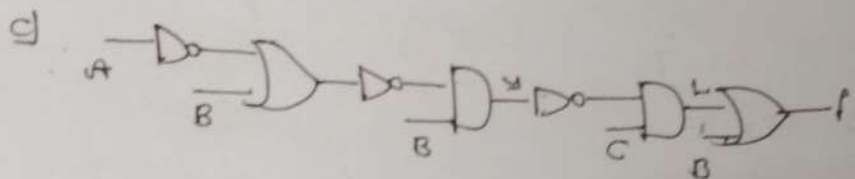
$$= AB$$

$$f = A + B$$

$$M = \bar{A}\bar{B} \cdot C$$

$$f = \bar{A}\bar{B} \cdot C + B$$

$$f = \underline{\underline{\bar{A}C + B}}$$



$$\bar{A} + B$$

$$= \overline{\bar{A} + B} = \bar{\bar{A}} \cdot \bar{B} = A \cdot \bar{B}$$

$$\Rightarrow Y = A \cdot \bar{B} \cdot B$$

$$Y = 0$$

$$\bar{Y} = 1$$

$$Y = 1 \cdot C = C$$

$$f = \underline{\underline{C + B}}$$

Q) Minimize the following Boolean expression using Boolean Postulates

$$(a) \bar{A}C\bar{D}E + \bar{A}\bar{B}\bar{D} + ABCE + AB\bar{D}$$

$$\bar{A}C\bar{D}E(\bar{B} + B) + \bar{A}\bar{B}\bar{D}$$

$$\underline{\underline{(\bar{A}C\bar{D}E + \bar{A}\bar{B}\bar{D} + AB(CE + \bar{D}))}}$$

$$\underline{\underline{(\bar{A}\bar{D}(CE + \bar{B}) + AB(CE + \bar{D}))}}$$

2) Minimize the following Boolean expression using Boolean Postulates

$$g) \bar{A}C\bar{D}E(B+\bar{B}) + \bar{A}\bar{B}\bar{D}(C+\bar{C})(E+\bar{E}) + ABCE(D+\bar{D}) + ABD(\bar{C}+\bar{C})(E+\bar{E})$$

$$\Rightarrow \bar{A}C\bar{D}EB + \bar{A}C\bar{D}E\bar{B} + \bar{A}\bar{B}\bar{D}CE + \bar{A}\bar{B}\bar{D}C\bar{E} + \bar{A}\bar{B}\bar{D}\bar{C}E + \bar{A}\bar{B}\bar{D}\bar{C}\bar{E} + ABCE\bar{D} + ABCE\bar{D} + ABCE\bar{D} + ABC\bar{D}E + ABC\bar{D}\bar{E} + ABC\bar{D}\bar{E} + ABD\bar{C}E + ABD\bar{C}\bar{E}$$

$$\Rightarrow \bar{A}BC\bar{D}E + \bar{A}\bar{B}C\bar{D}E + \bar{A}\bar{B}C\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}E + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E}$$

$$\bar{A}BC\bar{D}E + \bar{A}\bar{B}C\bar{D}E + \bar{A}\bar{B}C\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}E + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E}$$

$$\bar{A}BC\bar{D}E + \bar{A}\bar{B}C\bar{D}E + \bar{A}\bar{B}C\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}E + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E}$$

$$= ABC\bar{D}E(\bar{A}+A) + \bar{A}\bar{B}C\bar{D}(E+\bar{E}) + \bar{A}\bar{B}\bar{C}\bar{D}(E+\bar{E}) + \bar{A}\bar{B}\bar{C}\bar{D}(E+\bar{E}) + \bar{A}\bar{B}\bar{C}\bar{D}(E+\bar{E})$$

$$= ABC\bar{D}E + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D}$$

$$= BC\bar{D}E + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D}$$

$$= BC\bar{D}E + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C}$$

b) $\bar{A}\bar{B}\bar{C} + ABD + \bar{A}C + \bar{A}C\bar{D} + A\bar{C}D + A\bar{B}\bar{C}$

$\Rightarrow A\bar{B}\bar{C}(D+\bar{D}) + ABD(C+\bar{C}) + \bar{A}C(B+\bar{B})(D+\bar{D}) +$

$\bar{A}\bar{C}\bar{D}(B+\bar{B}) + A\bar{C}D(B+\bar{B}) + A\bar{B}\bar{C}(D+\bar{D})$

$\Rightarrow \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D} + ABD\bar{C} + ABD\bar{C} + \bar{A}CB\bar{D} + \bar{A}CB\bar{D} +$

$\bar{A}C\bar{B}D + \bar{A}C\bar{B}\bar{D} + \bar{A}C\bar{D}B + \bar{A}C\bar{D}\bar{B} + A\bar{C}DB +$

$A\bar{C}D\bar{B} + A\bar{B}\bar{C}D + A\bar{B}\bar{C}\bar{D}$

$\Rightarrow \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D} + ABD\bar{C} + A\bar{B}\bar{C}D + \bar{A}CB\bar{D} + \bar{A}CB\bar{D} +$

$\bar{A}C\bar{B}D + \bar{A}C\bar{B}\bar{D} + \bar{A}C\bar{D}B + A\bar{C}D\bar{B} + A\bar{B}\bar{C}D$

$\Rightarrow \bar{A}\bar{B}\bar{C}(D+\bar{D}) + ABD\bar{C} + \bar{A}CB\bar{C}(D+\bar{D}) + \bar{A}CB\bar{C}(D+\bar{D})$

$+ A\bar{B}\bar{C}(D+\bar{D}) + \bar{A}C\bar{D}B$

$\Rightarrow \bar{A}\bar{B}\bar{C} + ABD + \bar{A}CB + \bar{A}CB + A\bar{B}\bar{C} + \bar{A}C\bar{D}B$

$= \bar{B}\bar{C}(A+\bar{A}) + \bar{A}C(B+\bar{B}) + ABD + \bar{A}C\bar{D}B$

$= \bar{B}\bar{C} + \bar{A}C + ABD + \bar{A}C\bar{D}B$

$= \bar{B}\bar{C} + ABD + \bar{A}C(1+\bar{D}B)$

$\Rightarrow \bar{B}\bar{C} + ABD + \bar{A}C$

c) $\bar{A}\bar{C} + BC + A\bar{B} + \bar{A}BC + A\bar{C}\bar{D} + \bar{B}\bar{C}\bar{D}$

$\Rightarrow \bar{A}\bar{C}(B+\bar{B})(D+\bar{D}) + BC(A+\bar{A})(D+\bar{D}) + A\bar{B}(C+\bar{C})(D+\bar{D})$

$+ \bar{A}BC(D+\bar{D}) + A\bar{C}\bar{D}(B+\bar{B}) + \bar{B}\bar{C}\bar{D}(A+\bar{A})$

$= A\bar{C}BD + A\bar{C}B\bar{D} + A\bar{C}\bar{B}D + A\bar{C}\bar{B}\bar{D} + ABCD + ABC\bar{D} +$

$\bar{A}BCD + \bar{A}BC\bar{D} + A\bar{B}CD + A\bar{B}\bar{C}D + A\bar{B}\bar{C}\bar{D} +$

$A\bar{B}\bar{C}\bar{D} + \bar{A}BCD + \bar{A}BC\bar{D} + A\bar{B}CD + A\bar{B}\bar{C}D + A\bar{B}\bar{C}\bar{D} +$

$$\Rightarrow A\bar{C}BD + AB\bar{C}\bar{D} + A\bar{C}\bar{B}D + A\bar{B}\bar{C}\bar{D} + ABCD + ABCD +$$

$$\bar{A}BCD + \bar{A}BC\bar{D} + A\bar{B}CD + A\bar{B}C\bar{D} + A\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D}$$

$$\Rightarrow ABD(C+\bar{C}) + AB\bar{D}(C+\bar{C}) + A\bar{B}\bar{C}(D+\bar{D}) + \bar{A}BC(D+\bar{D})$$

$$+ A\bar{B}C(D+\bar{D}) + \bar{B}\bar{C}D(A+\bar{A})$$

$$= ABD + AB\bar{D} + A\bar{B}\bar{C} + \bar{A}BC + A\bar{B}C + \bar{B}\bar{C}D$$

$$\Rightarrow AB(D+\bar{D}) + A\bar{B}(\bar{C}+C) + \bar{A}B\bar{C} + \bar{B}\bar{C}D$$

$$= AB + A\bar{B} + \bar{A}BC + \bar{B}\bar{C}D$$

$$= A(B+\bar{B}) + \bar{A}BC + \bar{B}\bar{C}D$$

$$= A + \bar{A}BC + \bar{B}\bar{C}D$$

$$= A + BC + \bar{B}\bar{C}D$$

$$d) \bar{x}_1\bar{x}_3x_4 + x_3x_4 + \bar{x}_1\bar{x}_2x_4 + x_1x_2\bar{x}_3x_4 + x_2$$

$$\Rightarrow \bar{x}_1\bar{x}_3x_4 + x_3x_4 + \bar{x}_1\bar{x}_2x_4 + x_2(\bar{x}_1\bar{x}_3x_4 + 1)$$

$$\bar{x}_1\bar{x}_3x_4 + x_3x_4 + \bar{x}_1\bar{x}_2x_4 + x_2$$

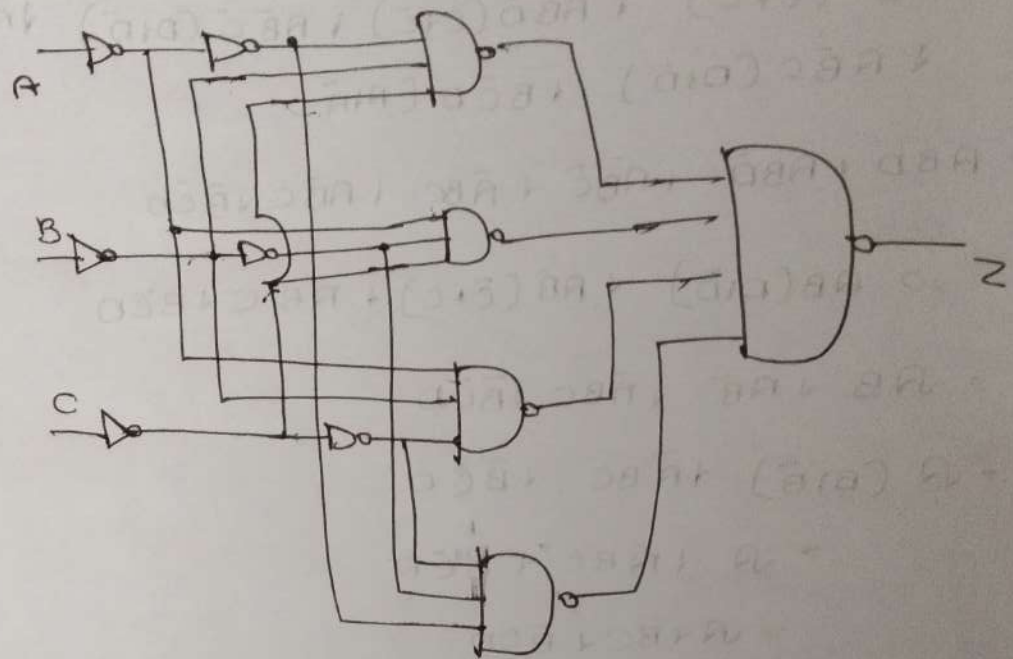
$$\bar{x}_1\bar{x}_3x_4 + x_3x_4 + x_2 + \bar{x}_1x_4$$

$$= \bar{x}_1\bar{x}_3x_4 + \bar{x}_1x_4 + x_3x_4 + x_2$$

$$\bar{x}_1(\bar{x}_3x_4 + x_4) + x_3x_4 + x_2$$

$$\bar{x}_1(\bar{x}_3 + 1)x_4 + x_3x_4 + x_2$$

3] Which single gate can represent the functionality of given circuit



$$Z_1: \overline{A} \overline{B} \overline{C} \Rightarrow \overline{A+B+C}$$

$$Z_2: \overline{A} B \overline{C} \Rightarrow A + \overline{B} + C$$

$$Z_3: \overline{A} \overline{B} C \Rightarrow A + B + \overline{C}$$

$$Z_4: A \cdot B \cdot C \Rightarrow \overline{A} + \overline{B} + \overline{C}$$

$$Z_2: \overline{(\overline{A} + B + C)} \cdot (A + \overline{B} + C) \cdot (A + B + \overline{C}) \cdot (\overline{A} + \overline{B} + \overline{C})$$

$$Z_2: \overline{(\overline{A} + B + C)} + (A + \overline{B} + C) + (A + B + \overline{C}) + (\overline{A} + \overline{B} + \overline{C})$$

$$Z_2: (A \overline{B} \overline{C}) + (\overline{A} B \overline{C}) + (\overline{A} \overline{B} C) + A B C$$

$$Z_2: A(\overline{B} \overline{C} + B C) + \overline{A}(B \overline{C} + \overline{B} C)$$

$$Z_2: A(B \oplus C) + \overline{A}(B \oplus C)$$

$$2 > A \oplus (B \oplus C)$$

$$2 > \cancel{A \oplus B} \oplus A \oplus B \oplus C$$

4) Minimize the following boolean expression using K-map

a. $f(a, b, c, d) = \sum m(4, 6, 8, 10, 11, 12, 15) + \sum (3, 5, 7, 9)$

cd \ ab	00	01	11	10
00			X	
01	1	X	X	1
11	1		1	
10	1	X	1	1

$$f = cd + \bar{a}b + a\bar{b} + a\bar{c}\bar{d}$$

b) $f(a, b, c, d) = \sum m(0, 2, 4, 6, 7, 9) + \sum d(10, 11)$

0			0
0		0	0
	0	X	X

$$f = (a+b) \cdot (a+\bar{b}+\bar{c}) \cdot (a+b+\bar{d})$$